Combined Science: Chemical Energetics

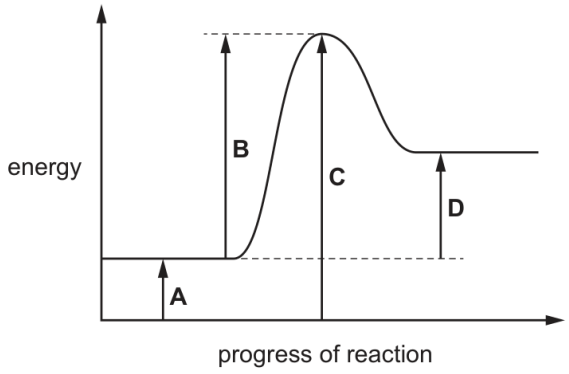
IGCSE Practice Paper

Total marks: 45 Estimated time: 54 minutes Questions: 18

Note: questions marked with * may stretch slightly beyond this chapter. They are included for stretch practice.

Write your answers in the boxes provided. Show working where appropriate.

Question 1. *[1 marks | Combined Science 2025 22]*



Question 2. *[1 marks | Combined Science 2025 12]*

19 Which reaction is **most** endothermic?

	initial temperature / °C	final temperature / °C
A	21	19
B	20	22
C	19	18
D	18	23

*Question 3. ** *[1 marks | Combined Science 2025 12]*

20 Which statement indicates that a chemical change occurs?

A Energy is absorbed.

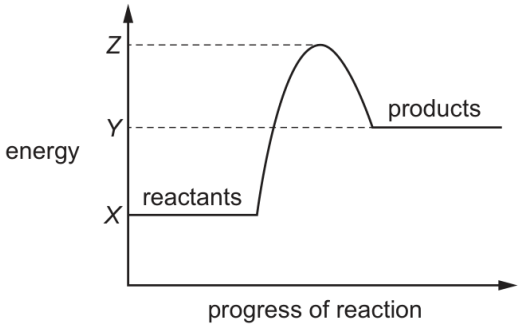
B There is a change in colour.

C There is a change of state.

D There is no change in mass.

Question 4. *[1 marks | Combined Science 2025 23]*

18 The reaction pathway diagram for a reaction is shown.



Which statement about the reaction is correct?

A The activation energy for this reaction is equal to the value of ($Z - Y$).

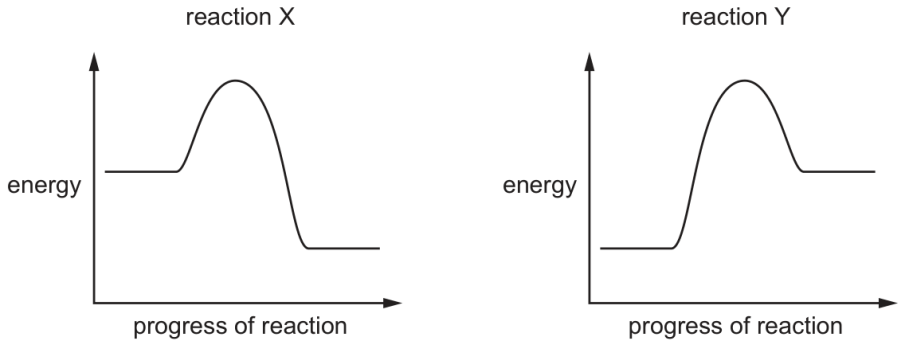
B The energy released by this reaction is equal to the value of ($Y - X$).

C The energy used to break bonds is more than the energy released in forming bonds.

D The overall energy change for this reaction is equal to the value of ($Z - X$).

Question 5. *[1 marks | Combined Science 2025 21]*

20 The reaction pathway diagrams for reaction X and for reaction Y are shown.



Which statement about the reactions is correct?

A Reaction X has a greater activation energy than reaction Y.

B Reaction X is endothermic and reaction Y is exothermic.

C The overall energy change in reaction X is much greater than in reaction Y.

D The temperature increases during reaction X and decreases during reaction Y.

Question 6. *[1 marks | Combined Science 2025 22]*

17 Which statement explains why a reaction is exothermic?

A More thermal energy is absorbed by breaking bonds than is released by making bonds.

B More thermal energy is absorbed by making bonds than is released by breaking bonds.

C More thermal energy is released by breaking bonds than is absorbed by making bonds.

D More thermal energy is released by making bonds than is absorbed by breaking bonds.

*Question 7. ** *[1 marks | Combined Science 2025 22]*

18 Which statement explains why the rate of a reaction is greater at a higher temperature?

A The activation energy is higher.

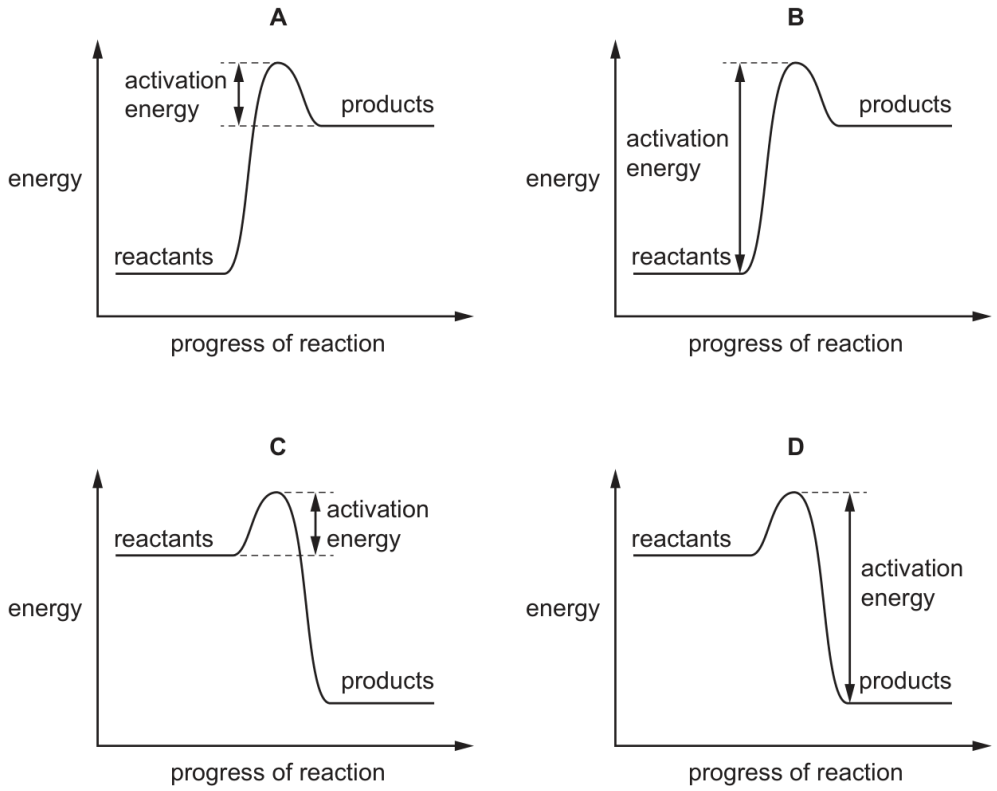
B More colliding particles have the minimum energy to react.

C The frequency of collisions between reacting particles decreases.

D There are more reacting particles per unit volume.

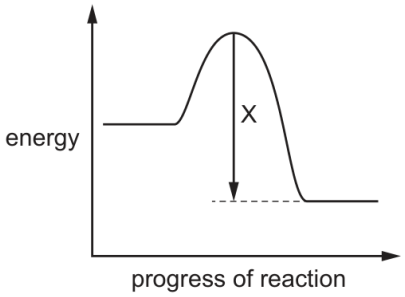
Question 8. *[1 marks | Combined Science 2025 22]*

18 Which reaction pathway diagram shows the activation energy for an exothermic reaction?



Question 9. *[1 marks | Combined Science 2025 21]*

15 A reaction pathway diagram for a reaction is shown.



Which statement describes and explains energy change X?

A Thermal energy is given out as bonds break.

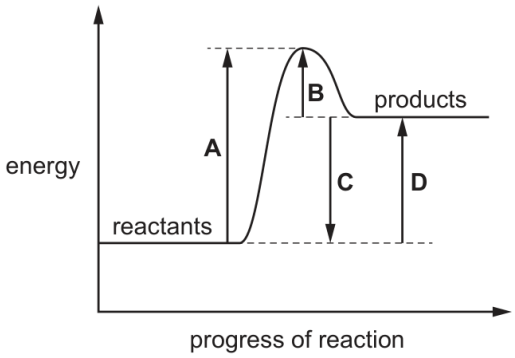
B Thermal energy is given out as bonds form.

C Thermal energy is taken in as bonds break.

D Thermal energy is taken in as bonds form.

Question 10. *[1 marks | Combined Science 2025 23]*

19 Which arrow on the reaction pathway diagram represents the overall energy change for the reaction?



Question 11. *[1 marks | Combined Science 2025 12]*

16 The initial and the final temperatures of four different reactions are recorded.

Which reaction is the **most** exothermic?

	initial temperature / °C	final temperature / °C
A	19	16
B	20	19
C	22	24
D	24	25

Question 12. *[1 marks | Combined Science 2025 13]*

16 The initial and the final temperatures of four different reactions are recorded.

Which reaction is the **most** exothermic?

	initial temperature / °C	final temperature / °C
A	19	16
B	20	19
C	22	24
D	24	25

*Question 13. ** *[3 marks | Combined Science 2025 31]*

6 **(a)** Fig. 6.1 shows the combustion of magnesium ribbon in air.

A white powder of magnesium oxide is formed.

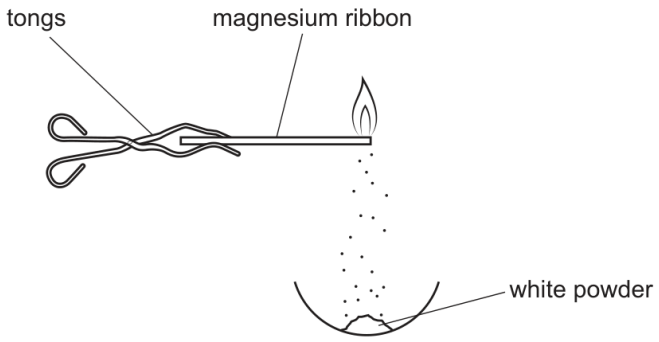


Fig. 6.1

(i) Name the gas that reacts with the magnesium.

..... [1]

(ii) Magnesium oxide is a metallic oxide.

Circle the term that describes this type of oxide.

acidic **aqueous** **basic** **carbonate** **solvent**

[1]

(iii) This reaction is exothermic.

Describe what is meant by exothermic.

.....

..... [1]

Question 14. *[3 marks | Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

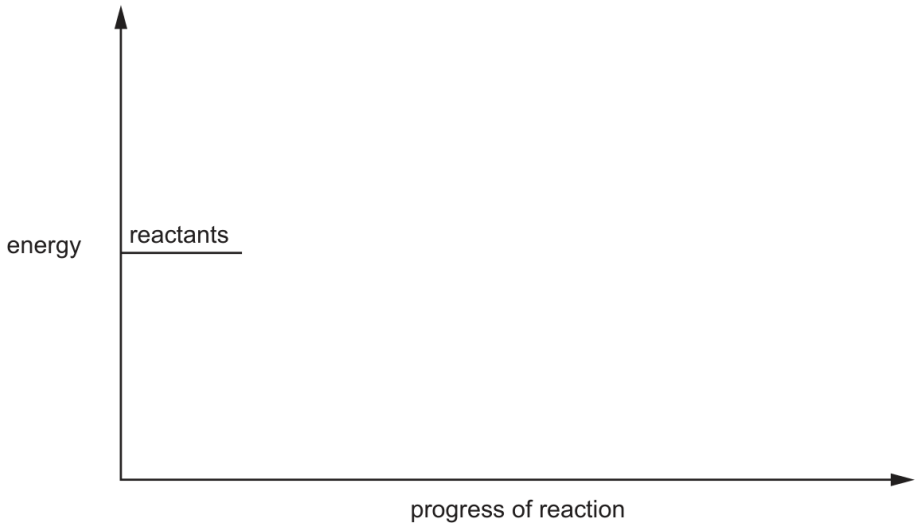


Fig. 5.1

[3]

[Total: 9]

Question 15. *[5 marks | Combined Science 2025 42]*

5 **(a)** Calcium oxide, CaO, is used to make cement.

Calcium oxide is formed by the decomposition of calcium carbonate, $CaCO_3$, at a very high temperature.

The equation for the reaction is shown.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

(i) Suggest why making cement contributes to global warming.

.....

.....

.....

..... [2]

(ii) The reaction is endothermic.

Explain what is meant by an endothermic reaction.

Use ideas about making bonds and breaking bonds in your answer.

.....

.....

.....

.....

..... [3]

Question 16. *[5 marks | Combined Science 2025 42]*

(iii) Fig. 5.1 shows the incomplete reaction pathway diagram for an endothermic reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

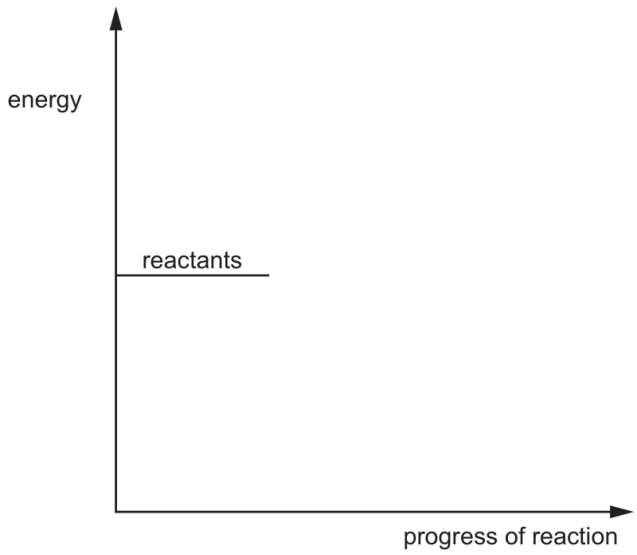


Fig. 5.1

[3]

(b) Calcium oxide reacts with water to form calcium hydroxide.

(i) The ions in calcium hydroxide are Ca^{2+} and OH^-.

Deduce the formula for calcium hydroxide.

..... [1]

(ii) Aqueous calcium hydroxide is an alkali.

State the colour of methyl orange indicator in aqueous calcium hydroxide.

..... [1]

[Total: 10]

Question 17. *[7 marks | Combined Science 2025 41]*

(c) Fig. 4.2 is a reaction pathway diagram for a reaction catalysed by copper.

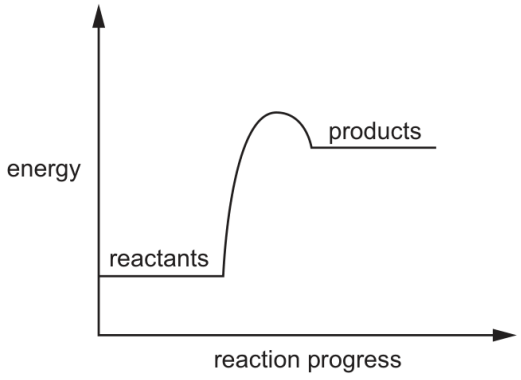


Fig. 4.2

(i) Identify the type of chemical reaction shown in Fig. 4.2.

Explain your answer.

type of chemical reaction .....

explanation .....

.....

..... [2]

(ii) Draw an arrow on Fig. 4.2 to show the activation energy for this reaction. [1]

(iii) Define activation energy.

.....

.....

.....

..... [2]

(iv) Describe what is meant by a catalyst.

.....

.....

.....

..... [2]

[Total: 11]

Question 18. *[10 marks | Combined Science 2025 42]*

(iii) Tick (✓) **all** the statements that are true for aqueous sodium hydroxide.

It is a covalent substance. ☐

It turns universal indicator purple. ☐

It reacts with dilute sulfuric acid to form sodium sulfate. ☐

It is a base but **not** an alkali. ☐

[2]

(b) The reaction between aqueous sodium hydroxide and dilute hydrochloric acid transfers thermal energy to the surroundings.

(i) Give **one** conclusion about the energy involved in bond breaking and bond making in this reaction.

.....

..... [1]

(ii) Complete Fig. 4.2 to show the reaction pathway diagram for this reaction.

Label:

- the reactants and products
- the overall energy change using an arrow.

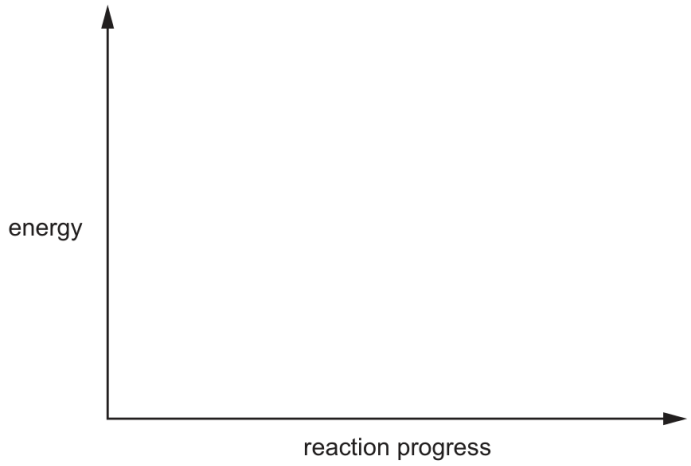


Fig. 4.2

[3]

[Total: 10]

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